DL midterm

Pravin patil

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(Q 11) Why single layer perceptron not compute XOR function?

→ Ans:-
Perceptron is the simplest neural unit. It is linearly classifier that can only learn linearly seperable patterns.

XOR is non linear function,
Truth table of XOR

0	1	0
0	1	1
1	0	1
1	1	0

No single straight line can separate the classes.
Hidden layer creates a new feature space where the problem becomes linearly separable.
[Input layer → Hidden layer → Output layer]
with 1 hidden layer of 2 units.
First lear NAND and of another AND them.

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(12) print o/p grad

a = 2.0

b = 3.0

y = a*b + b**2

y.backward()

$$y = ab + b^2$$

$$\frac{\partial y}{\partial a} = b$$

$$\frac{\partial y}{\partial b} = a + 2b$$

$$\boxed{\begin{array}{l} \text{output:} \\ a.grad = 3 \\ b.grad = 8 \end{array}}$$

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13) two layer feed forward network

$W_1 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ $w_2 = [1 \quad 1]$

$x = [1, 1, 1]^T$, $y = 50$

m.s.e loss?

$z = w \cdot x$

$= \begin{bmatrix} 1 + 2 + 3 \\ 4 + 5 + 6 \end{bmatrix} = \begin{bmatrix} 6 \\ 15 \end{bmatrix}$

$\bar{y} = 6 + 15 = 21$

$m.s.e = (\hat{y} - y)^2$

$= (21 - 50)^2$

$\boxed{m.s.e = 841}$

Q (14)

Transfer learning is a two phase training learning.

Phase 1 :- lets the new head learn without corrupting pre trained weights

phase 2: allows fine turing with stabe starting point.

In phase 1 it is freeze backbone, we can train only classified head. This allows model to train correctly without corrupting.
e.g. In mobileNetV2
we can set param.required_gradd = false.

Phase 2 :- we can unfreeze entire model.

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Q.15

conv1 (1, 8, k=3, s=1, p=1)

Relu

conv2 (8, 16, 3, 1, 1)

Relu

maxpool (2, 2)

→ I/P (1, 1, 28, 28)

$$O/P = \left[\frac{I - k + 2P}{Stride} \right] + 1$$

conv1 → (1, 8, 28, 28)

Relu 1 → (1, 8, 28, 28)

conv2 → 1, 16, 28, 28

maxpool → 14

output shape → (1, 16, 14, 14)

Total elements → 16 × 14 × 14
= 3136

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Q.16

Types of segmentation.
Semantic segmentation
instance segmentation
panoptic segmentation

	Semantic	instance	panoptic
(1) type of o/p	each pixel into category.	Detect each Object instance separately	Combines both into unified
	~~Detect~~ each Object		
Overlapping	cannot distinguish multiple instances	handles overlapping by masking	Overlapping by labelling
Representation	UNet		

Q.17

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Q.20

Conv1 (1, 32, 5, 1)
maxpool (2, 2)
Conv2 (32, 64, 5, 1)
maxpool (2, 2)

→ Input 28 × 28 $\left[\frac{I - K + 2P}{S}\right] + 1$

Conv1 (1 → 32, K = 5, S = 1, P = 0)
H -

H-out = floor $\left(\frac{28 - 5}{1}\right) + 1$

= 24

$\boxed{\text{Shape} = (1, 32, 24, 24)}$

~~Conv2 (1 → 32, K = 5, S = 1)~~

~~H out = floor(24 -~~

max pool = (1, 32, 24, 24)

$= \left[\frac{H - K}{S}\right] + 1$

$= \frac{\cancel{24} - 5}{1} + 1$

= ~~28~~ ~~20~~ 12

Max pool = ~~(1, 32, 28, 28)~~

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Q.20 continue

conv2 (1, 32, k=5, s=1)
(32 → 64, k=5, s=1)

$$H_{out} = \left[\frac{\cancel{28} - 5 + 0}{1} \right] + 1$$

(12 written above)

Hout = 24

~~(32, 64~~
~~(1, 32, 16, 16)~~

$\boxed{(1, 32, 9, 9)}$

maxpool(k=2, s=2)

= $\frac{24-2}{}$ (16 - 5) +1

= 12

$\boxed{O/P :- (1, 32, 19, 19)}$

Parameters:- 32 * 12 * 12 = 3708

(b) Linear expects → 64 * 7 * 7
= 3136

So, self. fc. will crash

(c) Total parameters :- 3708

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Q.21

RNN

$h_t = \tanh(W h_{t-1} + U x_t)$

(a)

$\frac{\partial h_t}{\partial h_1} = \sum_t \partial L_t / \partial W_{hh}$

(b) Gradient flows through multiple matrix multiplication (one per time steps)
long sequence →
many multiplications
→ long range dependencies.

LSTM

$c_t = f \odot c_{t-1} + i_t \odot c_t$

(a) c_t → solves vanishing gradient by separate memory highway that can store more information

Forget state → what to erase

as Sigmoid keep values in [0,1]
Thi

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Q.19

(a)

(b

Q.19

Softplus action:

$$sp(x) = \ln(1+e^x)$$

$$\text{sigmoid } \sigma(x) = \frac{1}{1+e^{-x}}$$

(a) $\Rightarrow$ calc. so

$$\frac{d}{dx}\left(\text{softplus}(x)\right)$$

$$= \frac{d}{dx}\left(\ln(1+e^x)\right)$$

$$= \cancel{\frac{d}{dx}\left(\frac{1}{1+e^x}\right)} \frac{d}{dx}\frac{1}{1+ex}$$

Hence $\frac{d}{dx}\left(\frac{1}{1+e^{-x}}\right) = \boxed{\frac{1}{1+e^{-x}}}$

Hence

$$\frac{d}{dx}\text{softplus}(x) = \sigma(x)$$

(b) $$\frac{d^2}{dx^2}\text{softplus}(x)$$

$$= \frac{d^2}{dx^2}\left(\ln(1+e^x)\right)$$

$$= \frac{d}{dx}\left(\frac{1}{1+e^x}\right)$$

$=$

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Q.18

$x = 2.0$

$w_1 = 3.0$

$w_2 = -1.0$

$b = 1.0$

(a) $z1 = w1 * x + b$

$= 3 * 2 + 1$

$\boxed{z_1 = 6}$

$a1 = relu(z_1) = max(0, 6) \Rightarrow 6$

$\boxed{a_1 = 6}$

$z2 = w_2 * a_1$

$= \cancel{(3) * 6}$ $(-1) * 6$

$z_2 = \cancel{18}$ -6

$loss = (z - y) * 2$ $(-6$

$= (\cancel{18} - 8) * 2$

$loss = 26$

(b) $x =$ torch.tensor (-5.0)

$z_1 = (3) \times (-5) + 1$

$= -15 + 1$

$= -14$

$a1 = Relu(z_1) = Relu(-14) = 0$

$z2 = w2 * a1$

$= -1.0$

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$Z_2 = w_2 * a_1$
$= (-1) * 6$
$= -6$

$loss (Z_1 - 5) * 2$
$= (-11) * 2$

$\boxed{loss = -22}$

$\boxed{x = 5.0}$

$Z_1 = (-5)(3) + 1$
$= -15 + 1$
$Z_1 = -14$

$Relu(Z_1) = \boxed{a_1 = 0}$